Lecture 4

Magnetism - Electromagnetism
Alternating Current (AC)

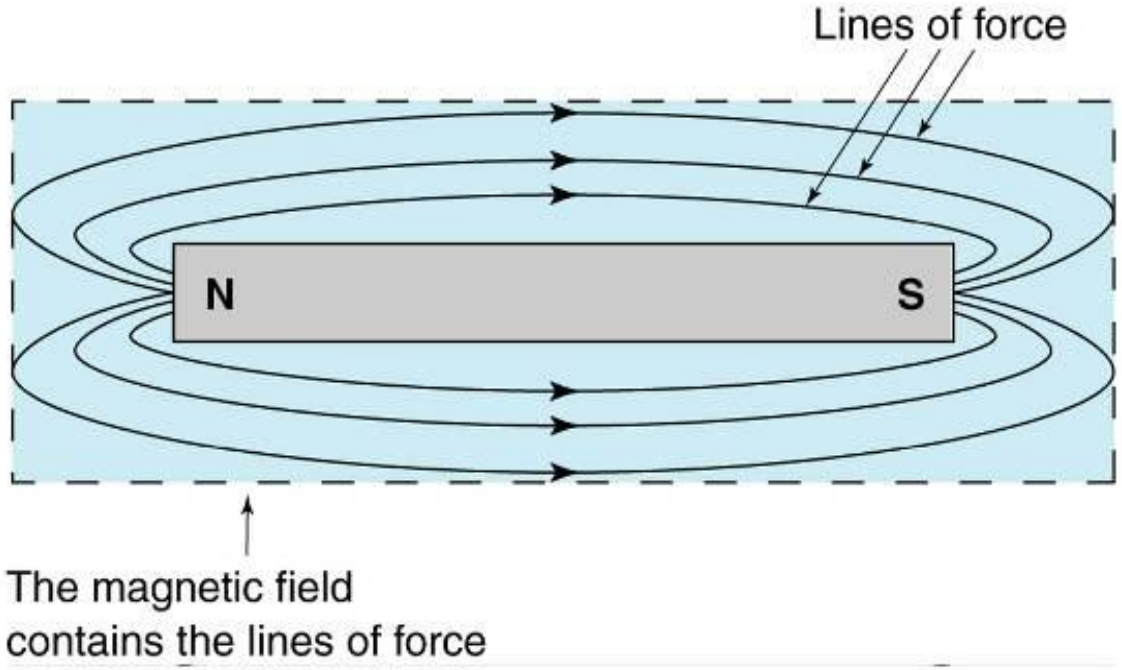
Slide Courtesy:
1.Electronics Technology Fundamentals, 3rd ed. Paynter and Boydell
2. Electronics Fundamentals 9th edition, Floyd/Buchla/Snyder

Outline

1. Magnetism
2. Electromagnetism
3. **Electromagnetic Induction**/Induced Voltage or Current
4. Alternating Current (AC) and Voltage
5. AC Circuit Analysis

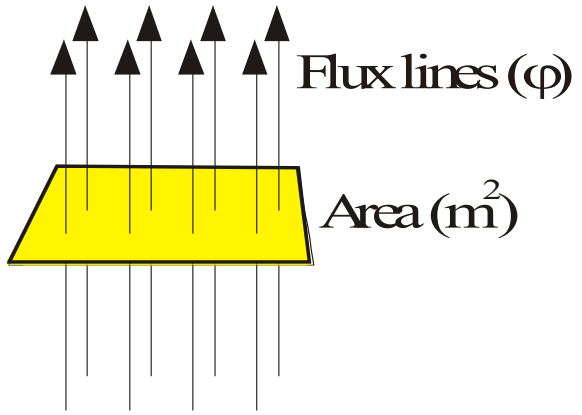
Magnetism – An overview

- Permanent magnet has **magnetic field** around it.
- Magnetic Field consists of **lines of force** that radiates from the north pole (M) to the south pole (S) and back to the N through the magnetic material.
- A group of lines of force is called **Magnetic Flux**



Magnetic Quantities

- **Magnetic Flux (φ):** the lines of force produced by a magnet
  - More lines of force, greater the value of flux
  - The unit of magnetic flux is the **weber (Wb)**, which is a measure of the total number of lines in a magnetic field.
  - 1 Wb $= 10^8$ lines



- **Magnetic Flux density (B):** The unit of **flux density** is the flux per unit area perpendicular to the magnetic field.
  - The unit of flux density is the **weber/square meter**, or **tesla** (T)

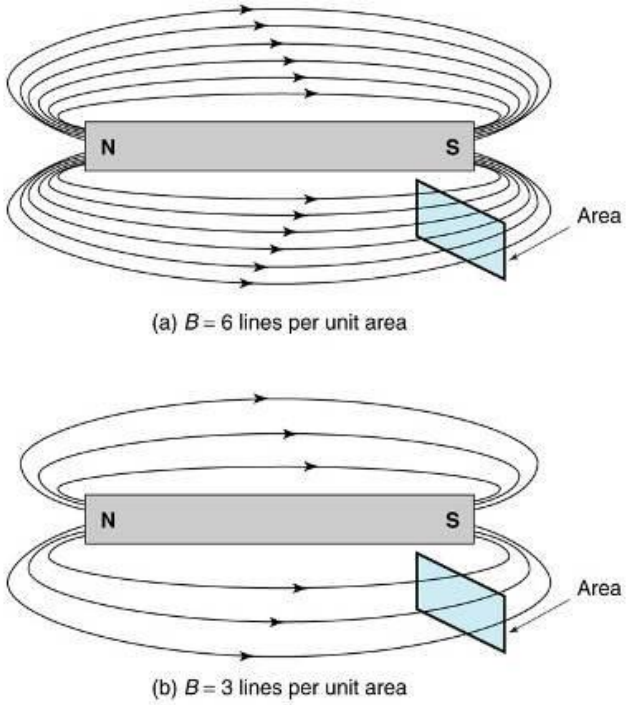
Flux density is given by the equation

$$B = \frac{\varphi}{A}$$

Where,

B = flux density (T)
φ = flux (Wb)
A = area (m^2)

Magnetic Quantities – Flux Density



Q.What is the flux density in a rectangular core that is 8.0 mm by 5.0 mm if the flux is 20 μWb?

Solution:

$$B = \frac{\varphi}{A} = \frac{20 \times 10^{-6}\ \text{Wb}}{(8.0 \times 10^{-3}\ \text{m})(5.0 \times 10^{-3}\ \text{m})}$$

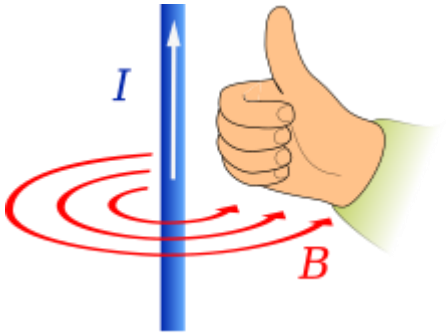
$$= 0.50\ \text{Wb/m}^2 =$$

0.5 T

Electromagnetism

- Electromagnetism is the production of a magnetic field by current in a conductor
- When an electric current flows through a conductor, it generates a magnetic field around it.
- Current-carrying wire generates a magnetic field around it is called **electromagnetic field**

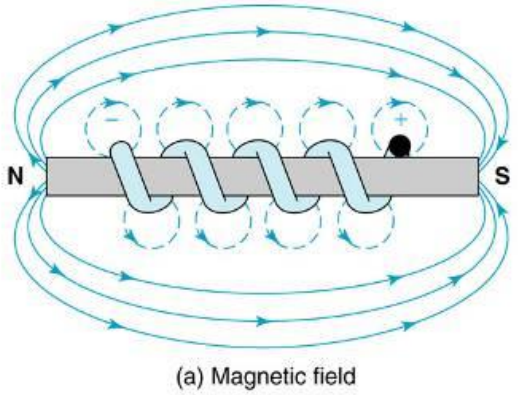
***Right-hand** screw rule*



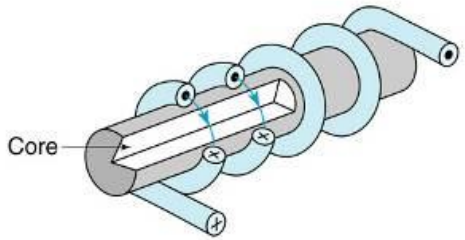
When the thumb is pointed in the **direction of conventional current** (from positive to negative),

Electromagnet - Magnetic Field Around a Coil

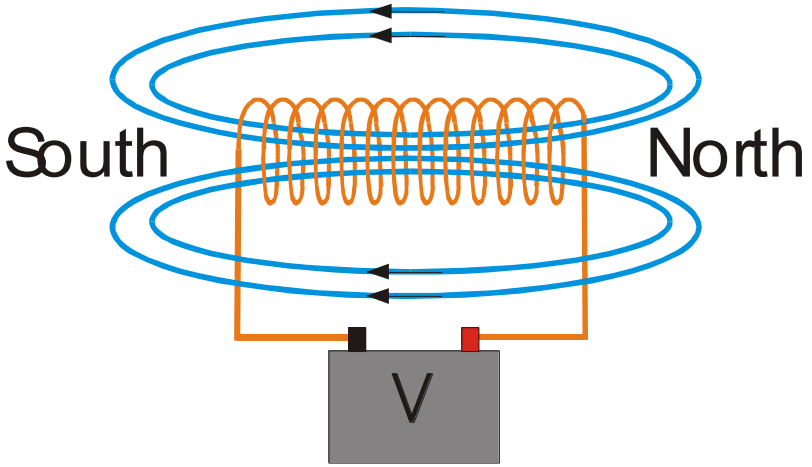
- The Coil – a wire wrapped into a series of loops for the purpose of concentrating magnetic lines of force
- Current in a coil causes a magnetic field. The coil acts as an electromagnet, with a north and south pole as in the case of a permanent magnet.



(a) Magnetic field



(b) Coil construction



Magnetic Quantities

Permeability (μ) defines the ease with which a magnetic field can be established in a given material. It is measured in units of the weber per ampere-turn meter.

The permeability of a vacuum (μ_0) is $4\pi \times 10^{-7}$ weber per ampere-turn meter, which is used as a reference.

Relative Permeability (μ_r) is the ratio of the absolute permeability to the permeability of a vacuum.

$$\mu_r = \frac{\mu}{\mu_0}$$

Magnetomotive Force (MMF)

Magnetomotive Force (MMF) is the force that produces magnetic flux in a magnetic circuit. It is generated by an electric current flowing through a coil of wire.

- Magnetic equivalent to electromotive force (EMF) - The cause of magnetic flux

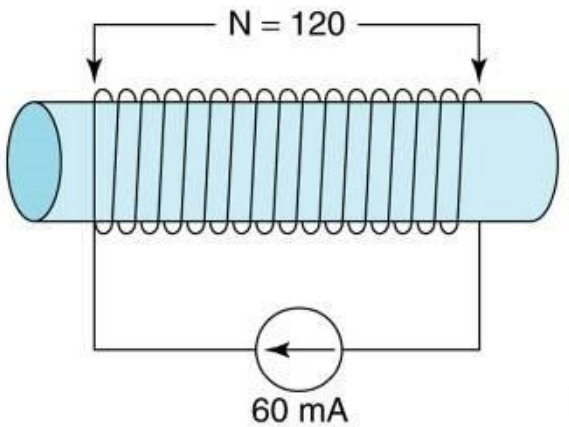
In an **electromagnet**, MMF is generated by the electric current passing through the coil.

A soft iron core inside the coil increases the magnetic flux by providing a low-resistance path for the magnetic field.

MMF is expressed in terms of **ampere-turns (At)**, which is the product of the current flowing through the wire and the number of turns (loops) in the coil.

$$\text{MMF} = NI$$

Where, N for number of turns, and I for current

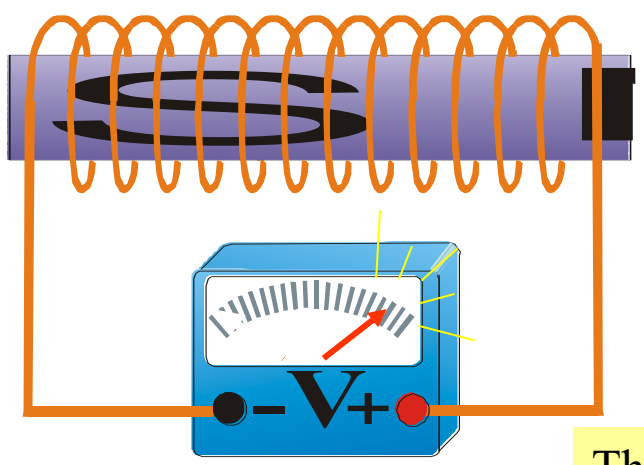


NI = (120)(60 mA)
= 7.2 A · t

Electromagnetic induction

- **Electromagnetic induction** is the process of generating an electric current in a conductor by changing the magnetic field around it.

Faraday's Law of Electromagnetic Induction:
1st Law: When the magnetic flux through a circuit changes, an electromotive force (EMF) is induced in the circuit.
2nd Law: The magnitude of the induced EMF is directly proportional to the rate of change of the magnetic flux.



Faraday experimented with generating current by relative motion between a magnet and a coil of wire. The amount of voltage induced across a coil is determined by two factors:

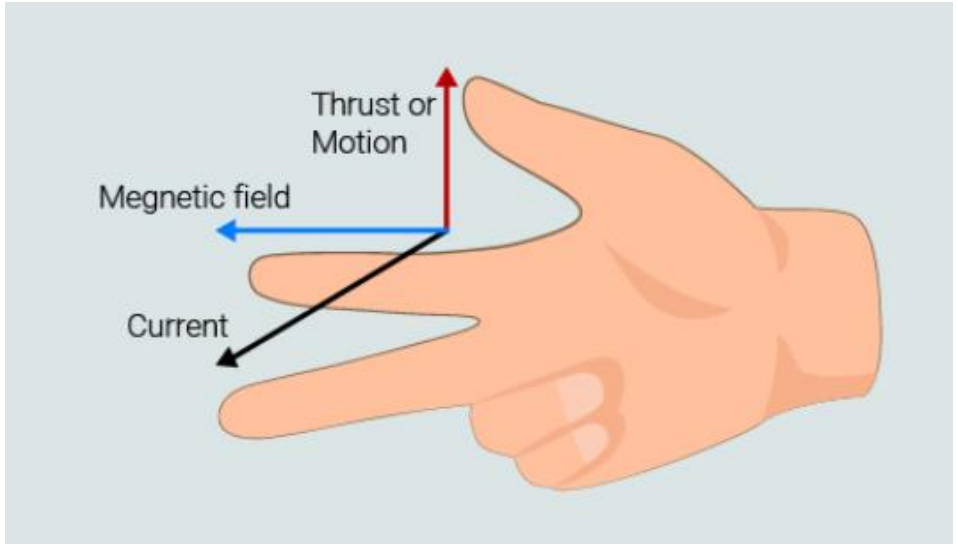
1. The rate of change of the magnetic flux with respect to the coil.
2. The number of turns of wire in the coil.

The induced voltage is greater when the coil has more turns.

Polarity of Induced Voltage

Fleming's Right-Hand Rule (generators)

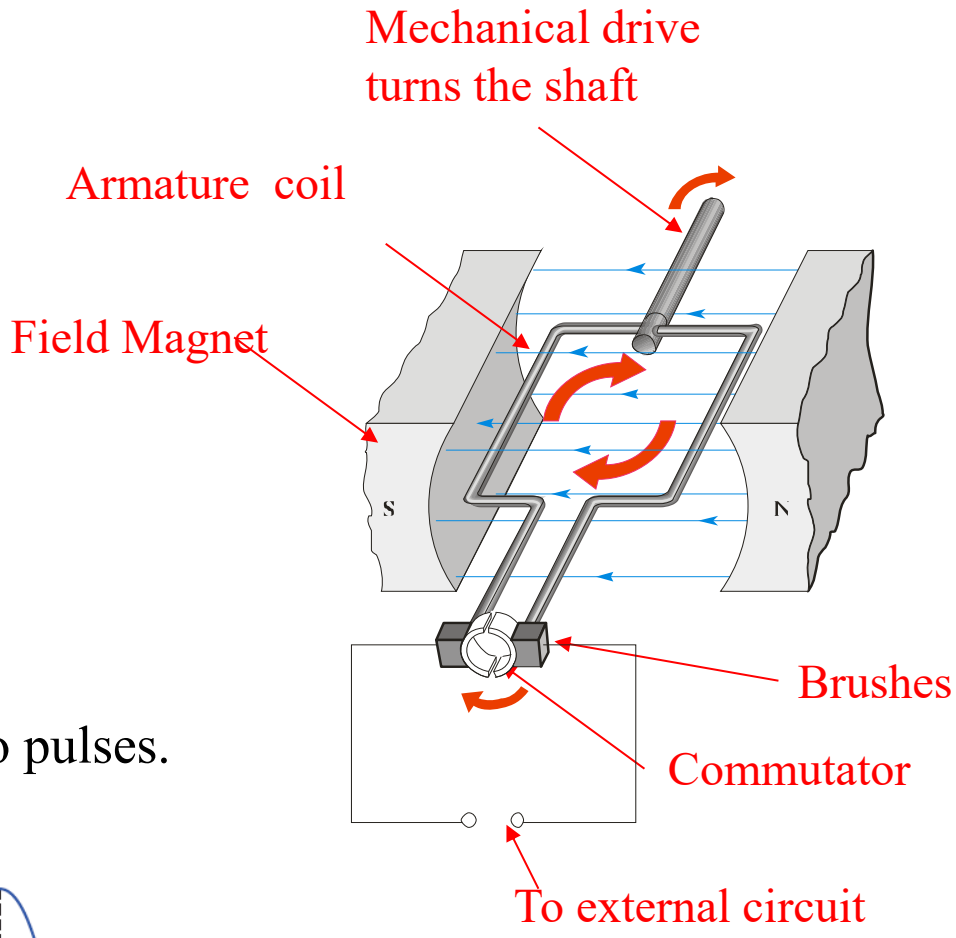
- Fleming's Right-Hand Rule is a practical tool used to determine the direction of induced current in a conductor moving through a magnetic field.
  - **Thumb:** Direction of **motion** of the conductor
  - **Forefinger:** Direction of the **magnetic field**
  - **Middle finger:** Direction of **induced current**



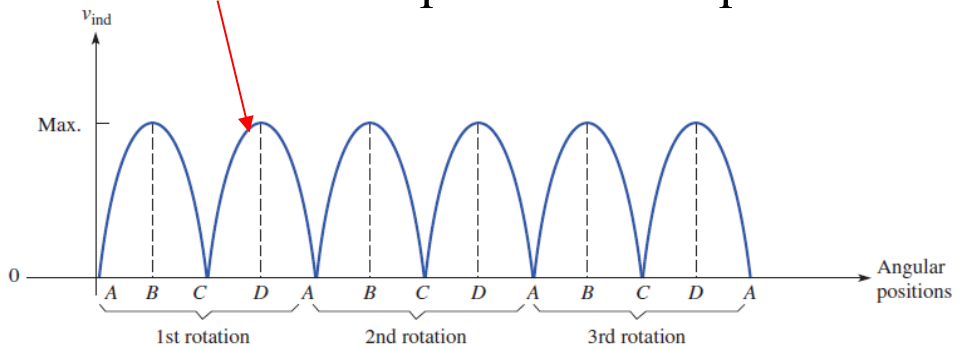
Application of Electromagnetic Induction - DC Generator

A **DC generator** (Direct Current generator) is a device that converts mechanical energy into electrical energy, producing direct current (DC). It operates on the principle of **electromagnetic induction.**

- **Armature** rotates within the magnetic field, it cuts through the magnetic flux, inducing an **electromotive force (EMF)** in the conductors.
- The induced EMF causes current to flow in the armature
- The **commutator** is a **split ring** connected to the ends of the armature winding. The commutator reverses the connection of the armature coil to the external circuit every half turn.
- This reversal ensures that the **current flows in one direction** in the external circuit, converting the internally generated AC into DC.
- The commutator effectively turns the alternating current into **pulsating direct current.**



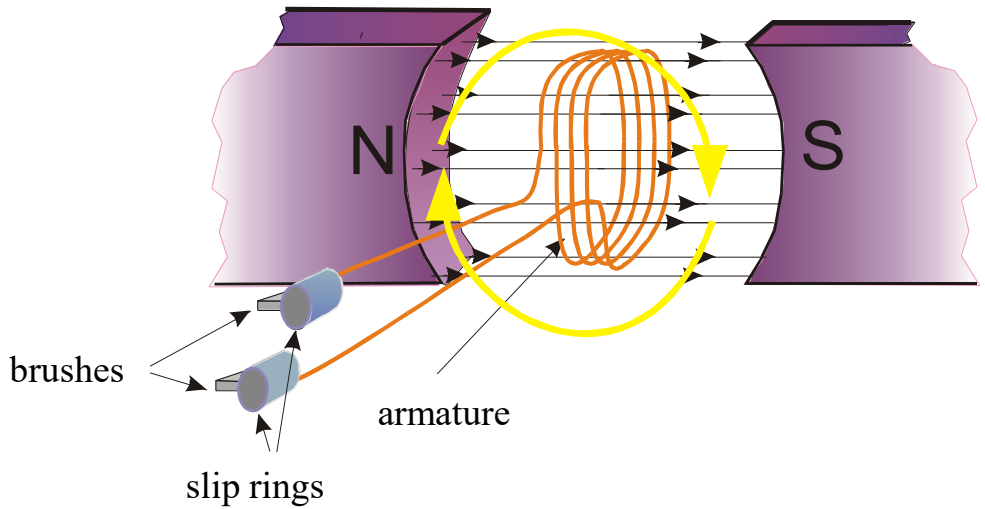
Each rotation produces two pulses.



AC Generator (Alternator)

- As the **armature** rotates, it cuts through the magnetic flux lines from the field magnet. This motion induces an **electromotive force (EMF)** in the armature windings.
- The direction of the induced EMF changes periodically, resulting in an **alternating current**
- The armature has an induced voltage, which is connected through **slip rings** and brushes to a load.
- The **slip rings** are continuous and rotate with the armature. Unlike the **commutator** in a DC generator, which reverses the connection every half turn, slip rings allow the alternating current to pass directly into the external circuit.

Small alternators may use a permanent magnet as shown here; other use field coils to produce the magnetic flux.



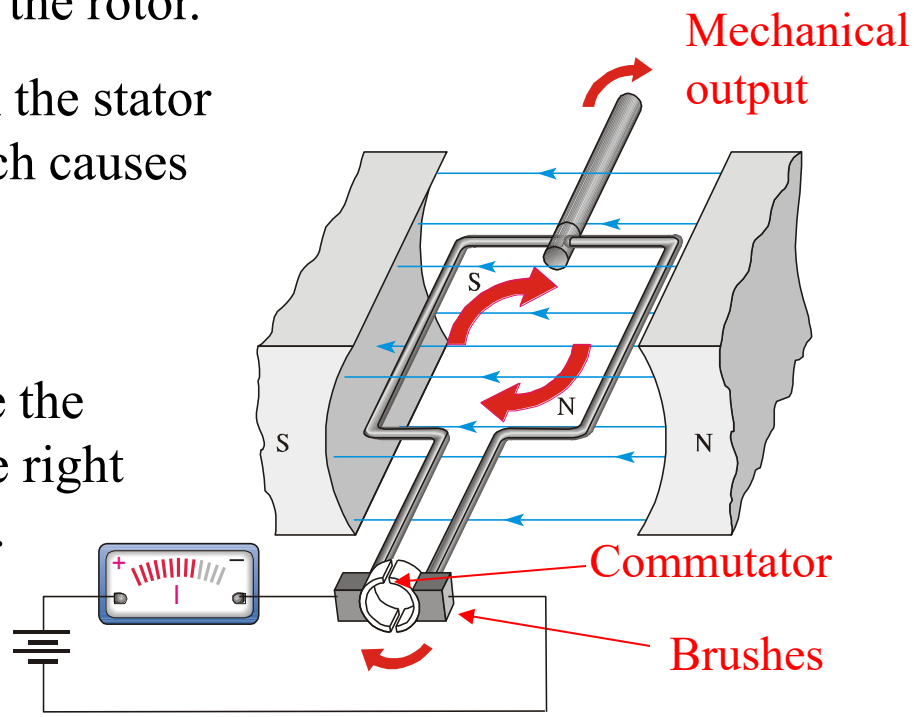
Electric generator (A.C. & D.C.) - video

DC Motor

A dc motor converts electrical energy to mechanical motion by action of a magnetic field set up by the rotor.

The rotor field interacts with the stator field, producing torque, which causes the output shaft to rotate.

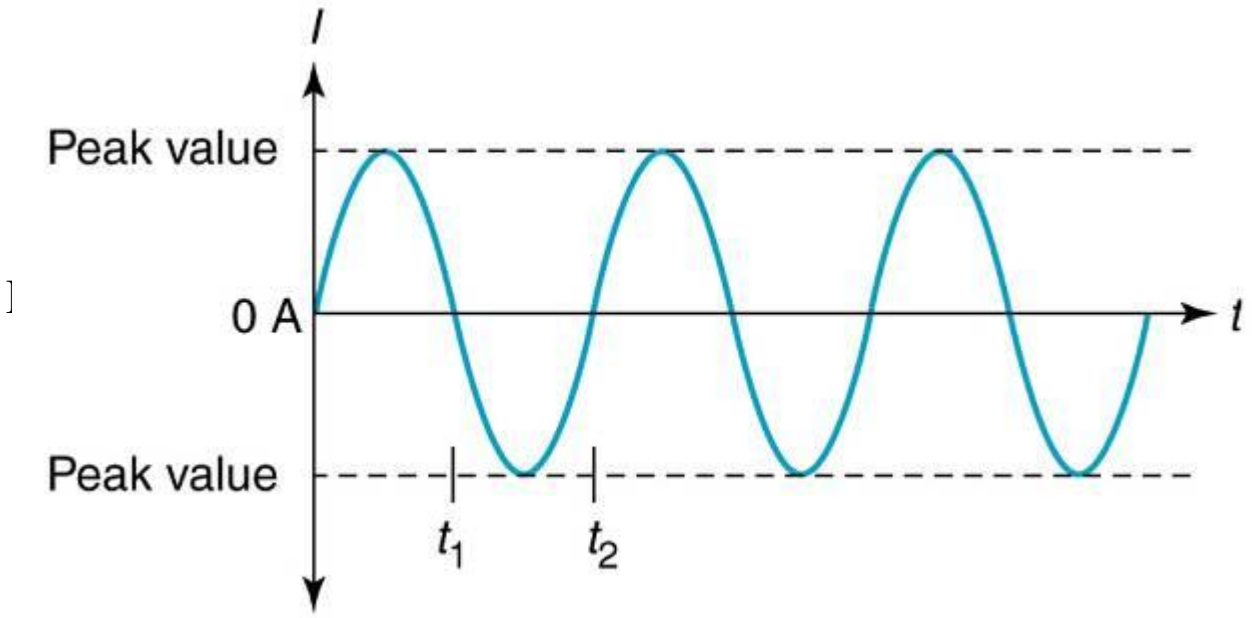
The commutator serves as a mechanical switch to reverse the current to the rotor at just the right time to continue the rotation.



Alternating Current (AC)

Alternating Current (AC)

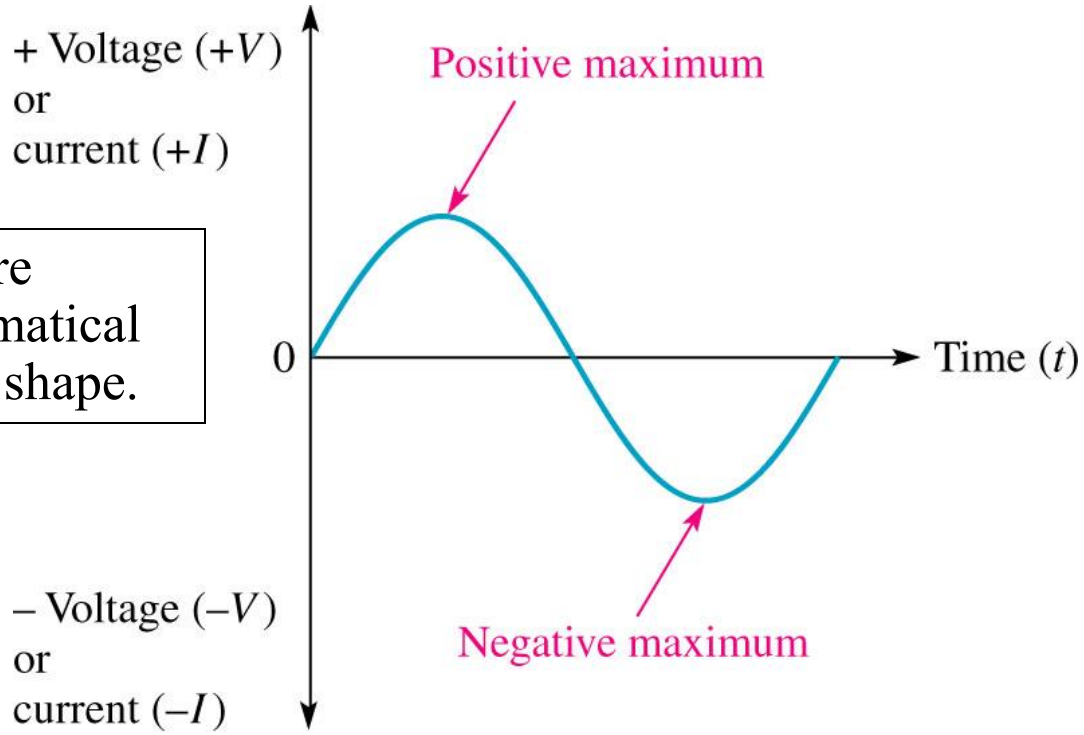
- Alternating Current – any current that periodically changes direction



Sine Waves

The sinusoidal waveform (sine wave) is the fundamental alternating current (ac) and alternating voltage waveform.

Electrical sine waves are named from the mathematical function with the same shape.



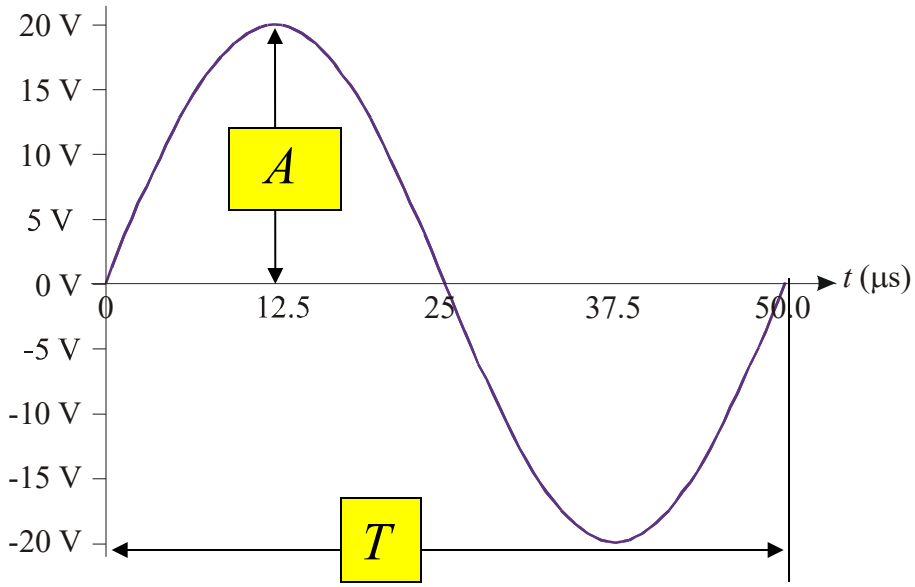
Amplitude and Period

Sine waves are characterized by the amplitude and period. The **amplitude** is the maximum value of a voltage or current measured from the mean; the **period** is the time interval for one complete cycle.

Example

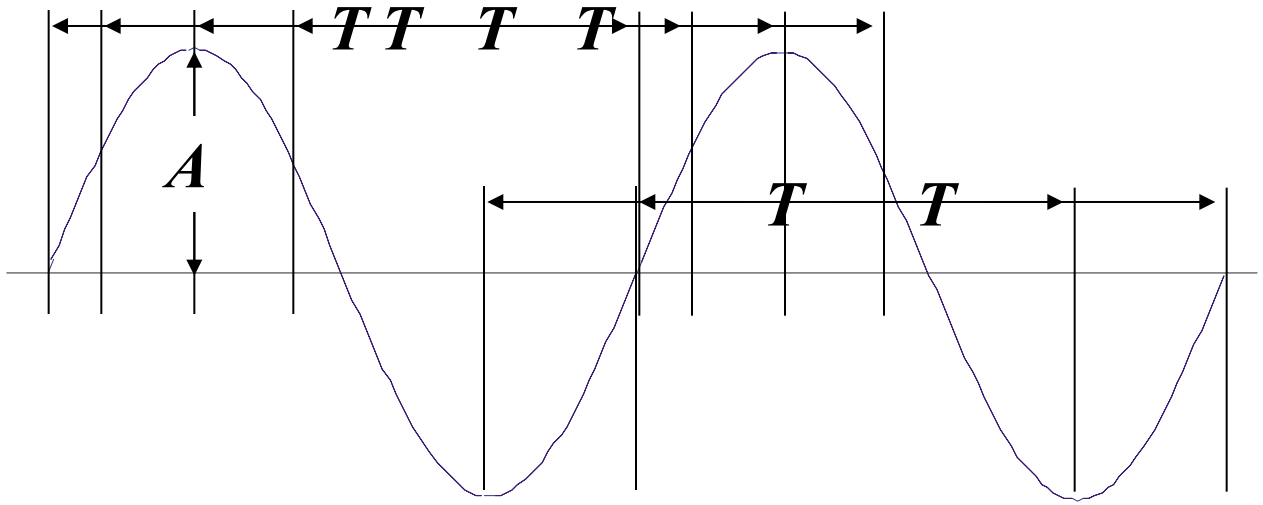
The amplitude (A) of this sine wave is 20 V

The period is 50.0 μs



Sine Waves

The period of a sine wave can be measured between any two corresponding points on the waveform.



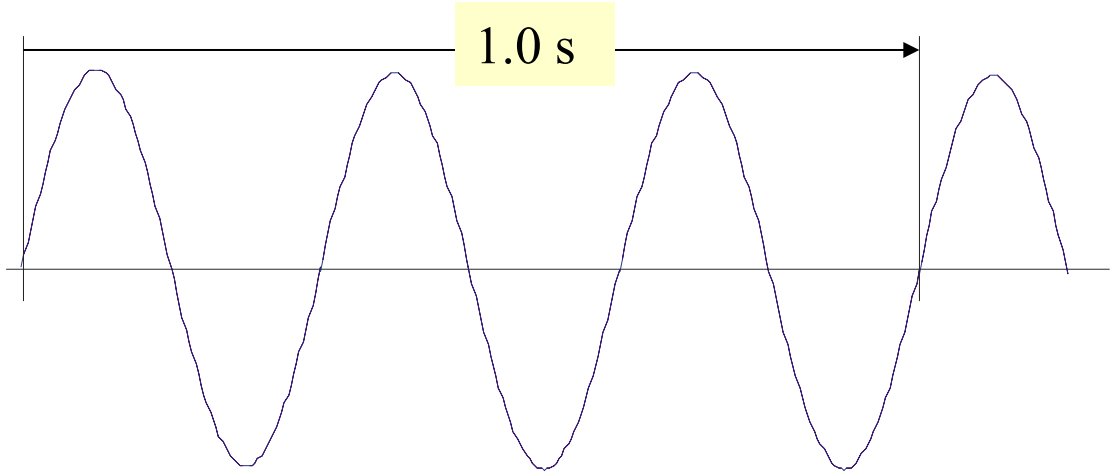
By contrast, the amplitude of a sine wave is only measured from the center to the maximum point.

Frequency

Frequency (f) is the number of cycles that a sine wave completes in one second. Frequency is measured in **hertz** (Hz)*.

Example

If 3 cycles of a wave occur in one second, the frequency is 3.0 Hz



*Frequency was originally measured in cycles per second (cps) but was changed to honor Heinrich Hertz.

Period and Frequency

- **Period** – the time required to complete one cycle of a signal
- **Frequency** – the rate at which the cycles repeat themselves
- The period and frequency are reciprocals of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Thus, if you know one, you can easily find the other.

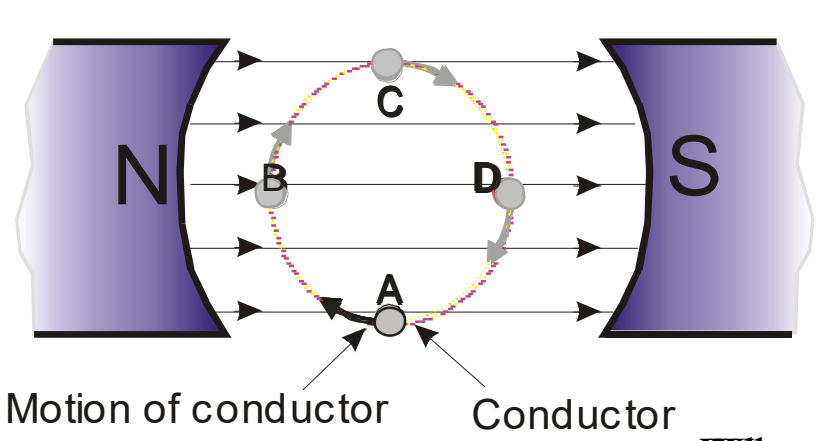
Example If the period is 50 μs, the frequency is

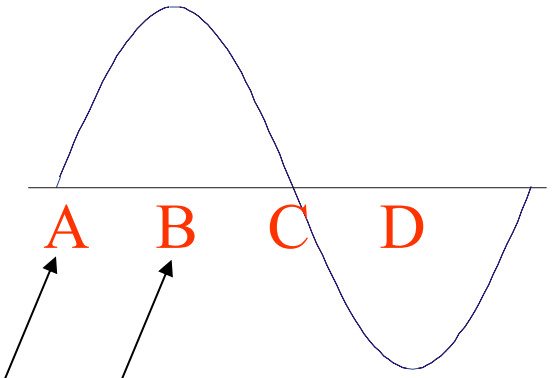
$$f = \frac{1}{T} = \frac{1}{50\ \mu s} = 20\ \text{kHz}$$

Generation of an Alternating Current

- **AC (Alternating Current)** is typically generated by **alternators** or **AC generators**. The process relies on the principle of **electromagnetic induction**, discovered by Michael Faraday.

When a conductor rotates in a constant magnetic field, a sinusoidal wave is generated.





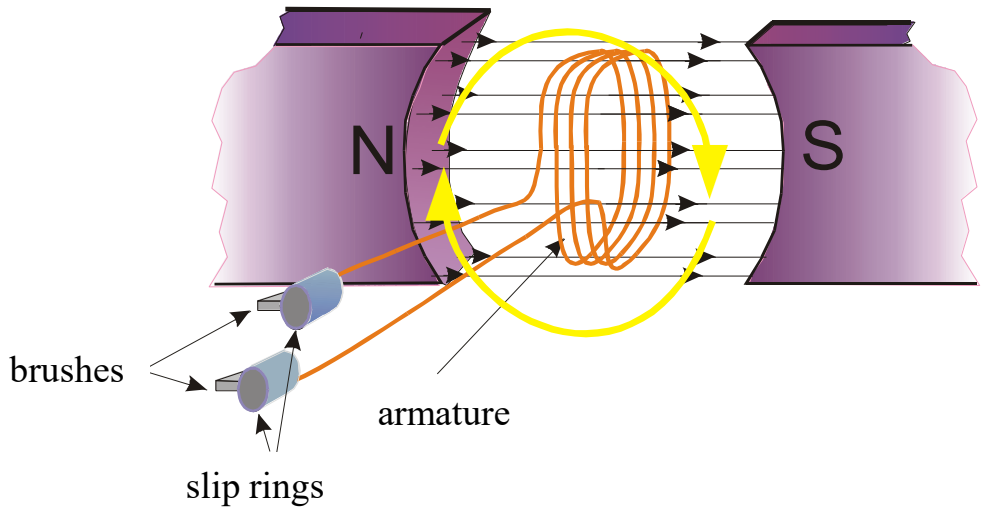
When the conductor is moving parallel with the lines of flux, no voltage is induced.

When the conductor is moving perpendicular to the lines of flux, the maximum voltage is induced.

AC Generator (Alternator)

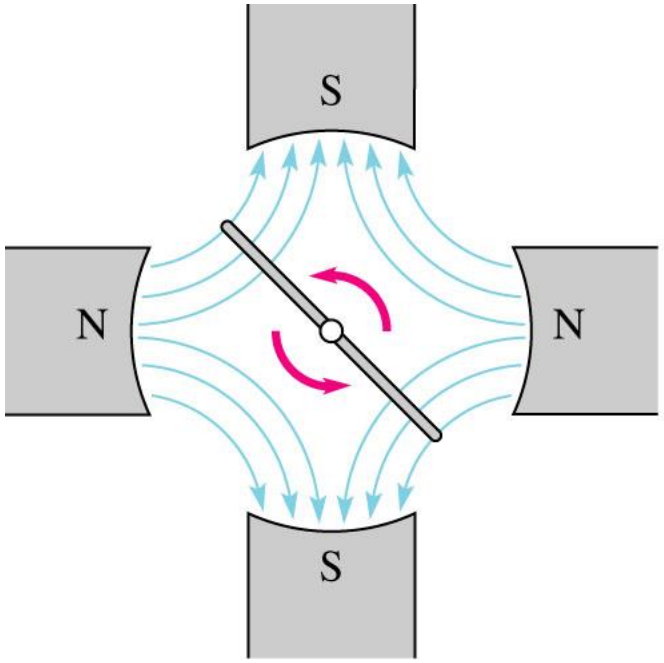
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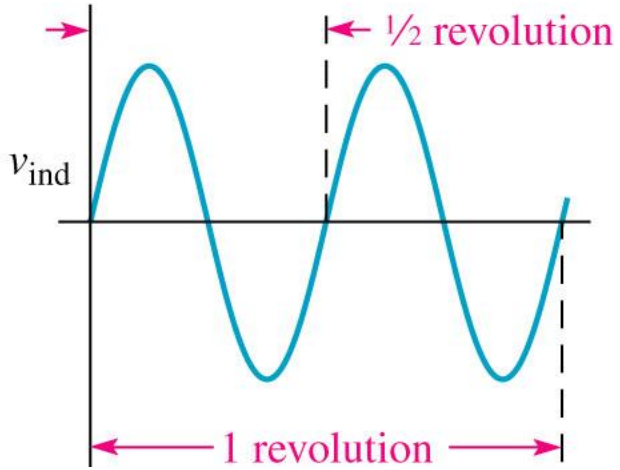
Small alternators may use a permanent magnet as shown here; other use field coils to produce the magnetic flux.



AC Generator (Alternator)

By increasing the number of poles, the number of cycles per revolution is increased. A four-pole generator will produce two complete cycles in each revolution.

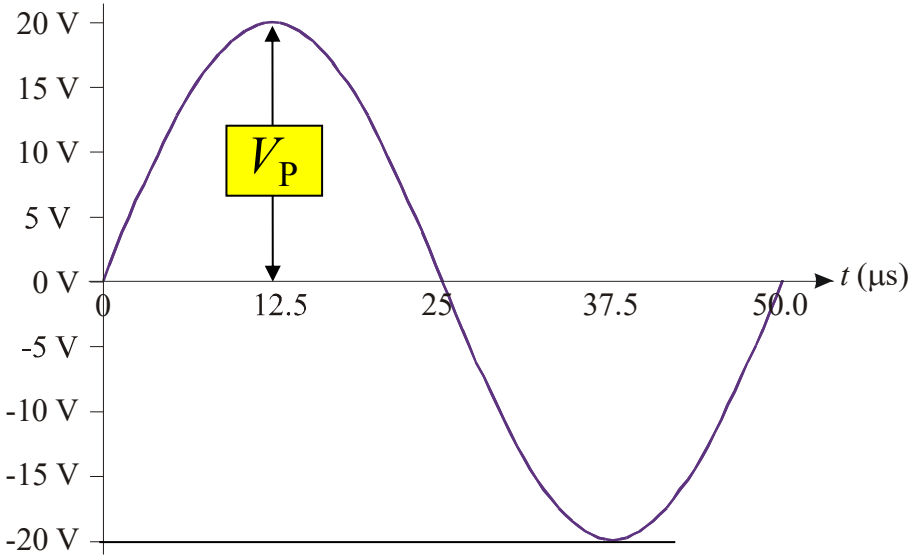




Measurements of a Sine Wave

There are several ways to specify the voltage of a sinusoidal voltage waveform.

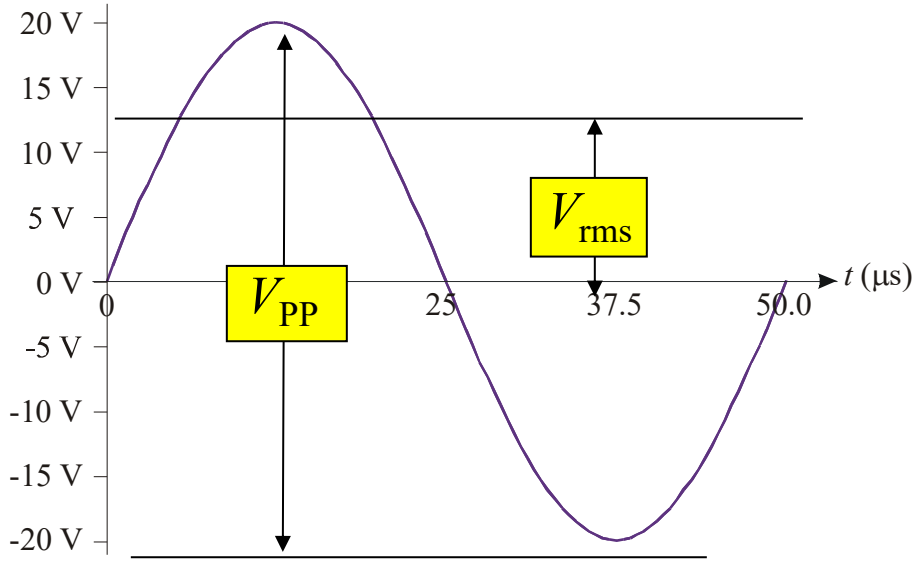
- **Peak voltage (Vp):** The amplitude of a sine wave is also called the peak value, abbreviated as V_P for a voltage waveform.
- **Peak-to-peak (Vpp):** The peak-to-peak is twice the peak value



The peak voltage of this waveform is 20 V.

RMS (root mean square) value (Vrms):

- The rms value is 0.707 times the peak value.
- The **rms value** of a sine wave gives the equivalent DC value that would deliver the same power to a resistor as the AC signal does.
- Multimeters always read rms values for ac voltage and current



The peak-to-peak voltage is 40 V_{pp}.

The rms voltage is 14.1 V_{rms}.

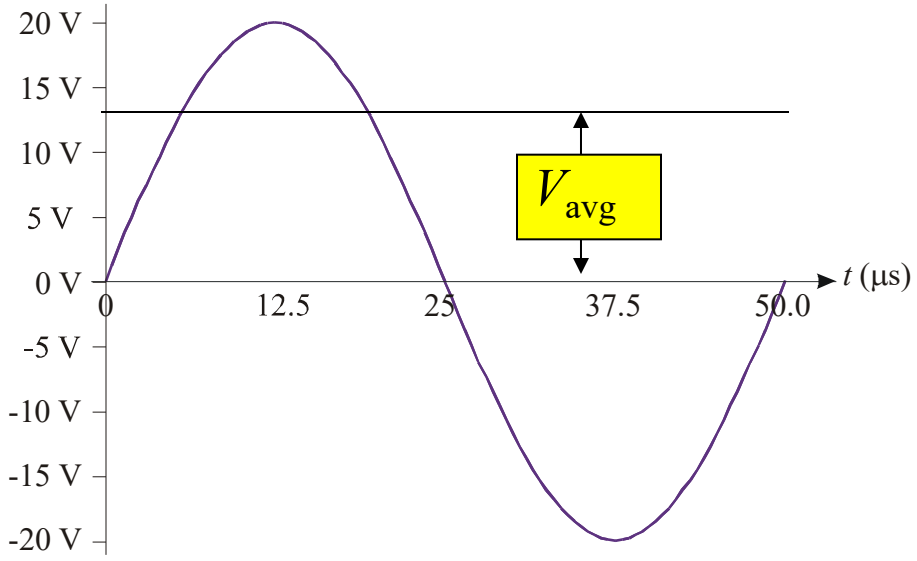
Average value (Vave): For some purposes, the average value (actually the half-wave average) is used to specify the voltage or current.

The average value is as 0.637 times the peak value.

Vave = 0.637* Vp

Example

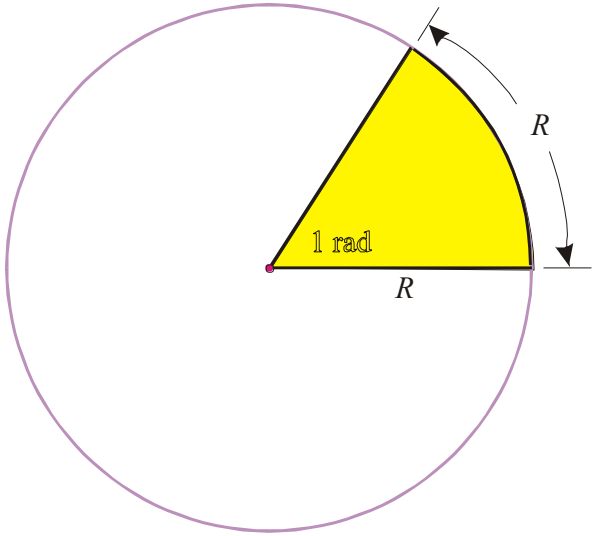
The average value for the sinusoidal voltage is 12.7 V.

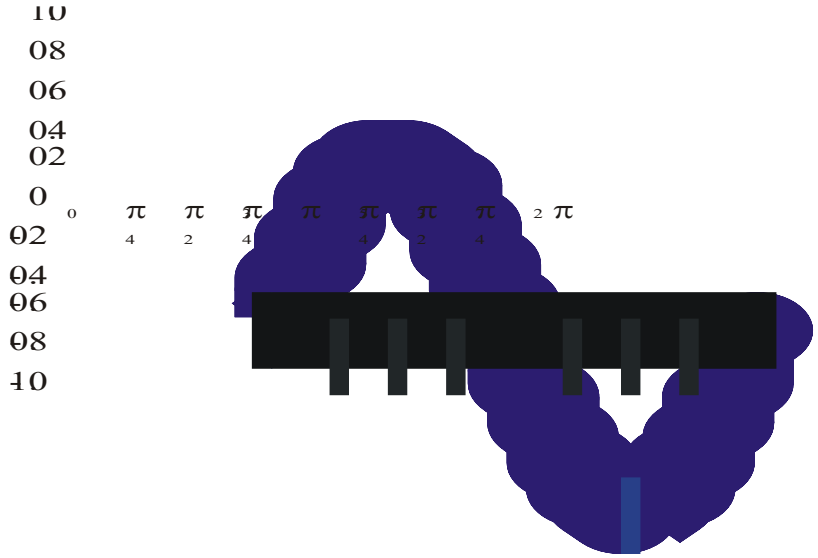


Angular Measurement

Angular measurements can be made in degrees (°) or radians. The radian (rad) is the angle that is formed when the arc is equal to the radius of a circle. There are 360° or 2π radians in one complete revolution.

$$\text{rad} = \frac{\pi \text{ rad}}{180°} \times \text{degrees}$$





Sine Wave Equation

Instantaneous values of a wave are shown as v or i. The equation for the instantaneous voltage (v) of a sine wave is

$$v = V_p \sin\theta$$

Where,

V_p = peak voltage

θ = angle in rad or degrees

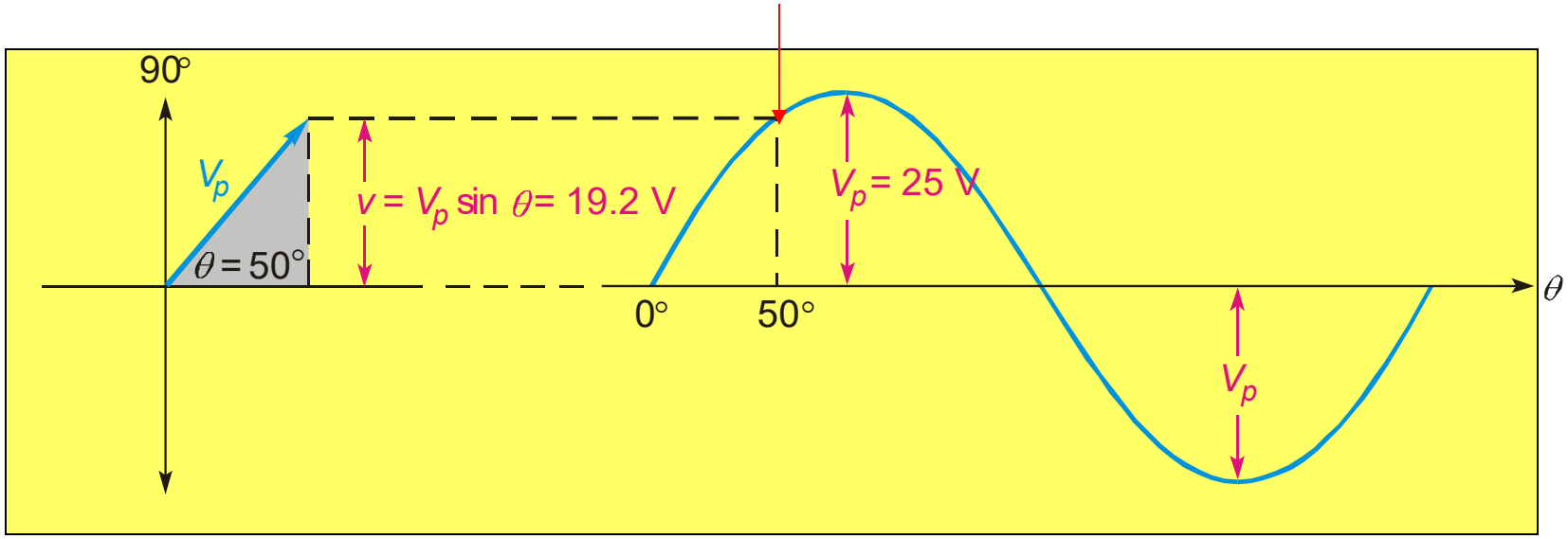
Example

If the peak voltage is 25 V, the instantaneous voltage at 50 degrees is 19.2 V

What is this angle in radians? 0.873 rad

Sine Wave Equation

A plot of the example in the previous slide (peak at 25 V) is shown. The instantaneous voltage at 50° (or 0.873 rad) is 19.2 V as previously calculated.



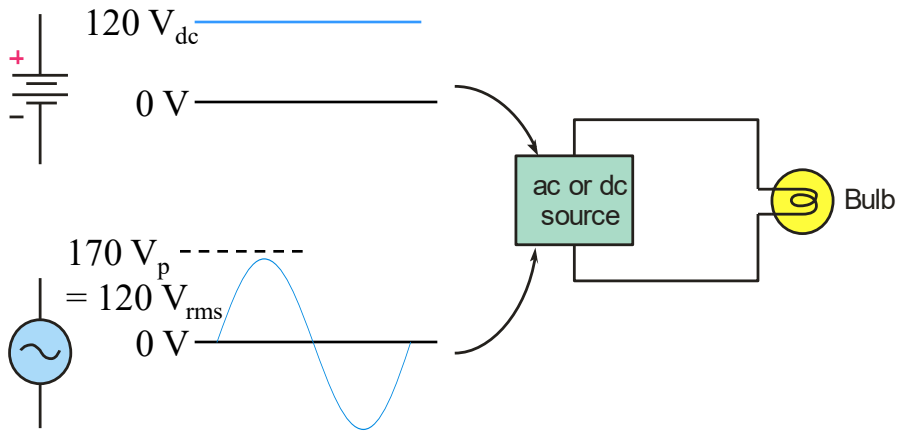
Analysis of AC Circuits

- Resistive ac circuits are analyzed with the same circuit laws and power formulas as for dc circuits including Ohm's law and Kirchhoff's laws. With ac circuits, voltage and current must be expressed consistently.
- **For power calculations, apply rms values**. When specifying ac as rms values, ac or dc sources will produce the same power to a given load.

$$P = V_{\text{rms}} I_{\text{rms}}$$

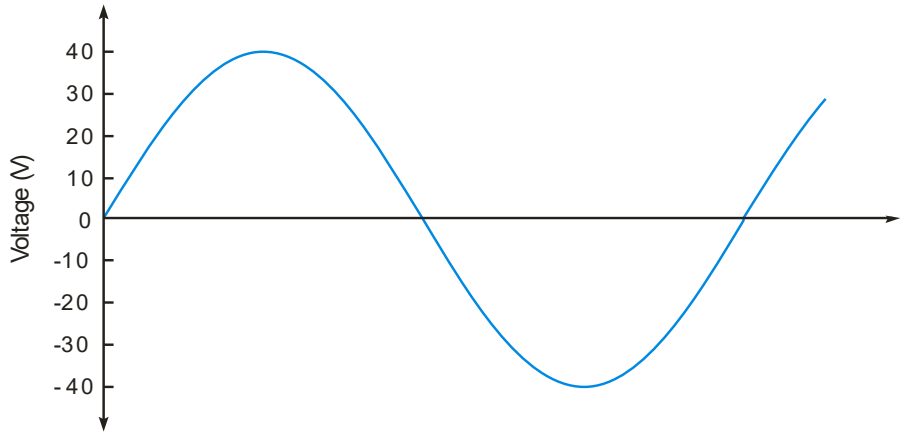
$$P = \frac{V_{rms}^2}{R}$$

$$P = I_{\text{rms}}^2 R$$



Analysis of AC Circuits

Example Assume a sine wave with a peak value of 40 V is applied to a 100 Ω resistive load. What power is dissipated?



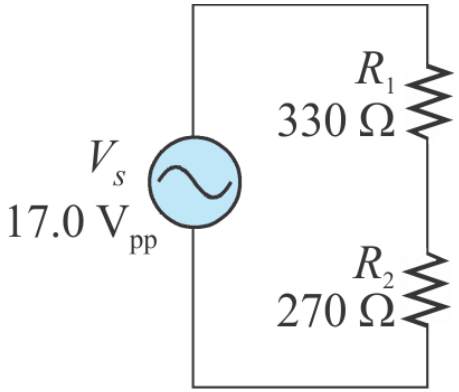
Solution

$$V_{rms} = 0.707 \times V_p = 0.707 \text{ x } 40 \text{ V} = 28.3 \text{ V}$$

$$P = \frac{V_{rms}^2}{R} = \frac{28.3 \text{ V}^2}{100\ \Omega} = 8.0 \text{ W}$$

Analysis of AC Circuits

Example Calculate the power in R_2.



To calculate power, you need to use rms values.
Converting V_S:

$$V_s = \frac{(17.0\ \mathrm{V_{pp}})}{2.828} = 6.01\ \mathrm{V_{rms}}$$

Use the voltage-divider rule to find V_2:

$$V_2 = V_S \frac{R_2}{(R_1 + R_2)} = 6.01\ \mathrm{V} \frac{270\ \Omega}{(330\ \Omega + 270\ \Omega)} = 2.71\ \mathrm{V}$$

Apply Watt's law to find P_2: $P_2 = \frac{V_2^2}{R_2} = \frac{(2.71\ \mathrm{V})^2}{270\ \Omega} =$ 27.1 mW

Some interesting readings

- AC vs DC war : War of the Currents
- Electric generator (A.C. & D.C.) - video

Next Class : Mid-term 1